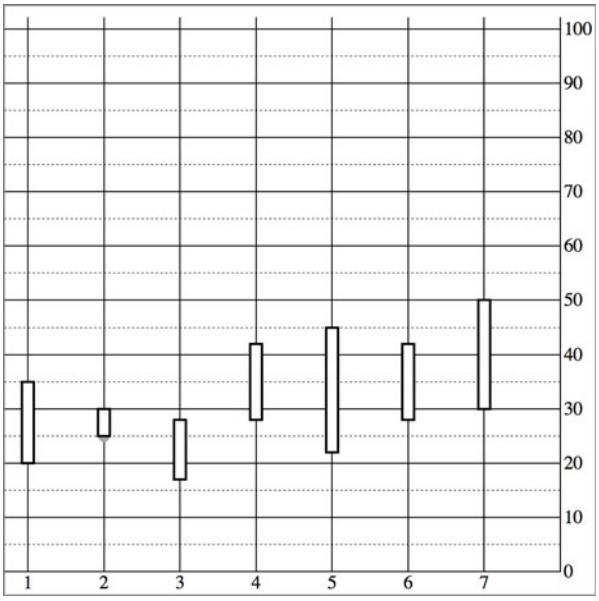


Figure 4.14 The `<rect>` elements are now properly placed so that their top and bottom correspond with the visitor age between the first and third quartiles of visitors for each day. The circles are completely covered, except for the second rectangle where the first quartile value is the same as the median age, and so we can see half the gray circle peeking out from underneath it.

```
…
.each(function(d,i) {
    d3.select(this)
        .append("rect")
        .attr("width", 20)
        .attr("x", -10)
        .attr("y", yScale(d.q3) - yScale(d.median))
        .attr("height", yScale(d.q1) - yScale(d.q3))
        .style("fill", "white")
        .style("stroke", "black");
});
```

Sets a negative offset of half the width to center a rectangle horizontally

The height of the rectangle is equal to the difference between its ql and q3 values, which means we need to offset the rectangle by the difference between the middle of the rectangle (the median) and the high end of the distribution—q3.

We'll use the same technique we used to create the chart in figure 4.14 to add the remaining elements of the boxplot (described in detail in figure 4.15) by including several append functions in the `.each()` function. They all select the parent `<g>` element created during the data-binding process and append the shapes necessary to build a boxplot.

Listing 4.5 The `.each()` function of the boxplot drawing five child elements

```
…
.each(function(d,i) {
    d3.select(this)
        .append("line")
        .attr("class", "range")
        .attr("x1", 0)
        .attr("x2", 0)
        .attr("y1", yScale(d.max) - yScale(d.median))
        .attr("y2", yScale(d.min) - yScale(d.median))
```

Draws the line from the min to the max value